Responsibility Attribution Prediction from Regulatory Text: A Linguistic Analysis of Accountability Construction

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Abstract Responsibility attribution in regulatory investigations is not discovered from evidence — it is linguistically constructed through word choice, syntactic framing, and discourse patterns. I present a text-based classification framework that predicts responsibility attribution (Firm, Individual, System) from violation text alone, using TF-IDF features and Logistic Regression. Applied to an annotated corpus of GMP regulatory language (n=45) with Leave-One-Out cross-validation, the framework achieves macro F1 = 0.725 and demonstrates that Individual-level attribution is most linguistically distinct (AUC = 0.940), while System-level attribution exhibits the lowest discriminability (AUC = 0.764), reflecting linguistic overlap with Firm-level language. When applied to the FDA Warning Letter corpus (N=50, 160 violations, 2016–2025), the model reassigns 49 of 160 Firm-labeled violations to System, suggesting that regulatory enforcement documents systematically attribute infrastructure failures to organizational responsibility. LIME analysis reveals class-specific linguistic markers: Firm attribution is signaled through systemic failure language (*inadequate*, *failure to establish*), Individual through agent-action language (*operator*, *analyst*), and System through infrastructure language (*equipment*, *malfunction*).

Keywords: responsibility attribution, regulatory NLP, text classification, accountability, FDA Warning Letters, LIME, linguistic analysis

Introduction

Accountability in regulated environments is not merely a legal outcome — it is produced through language. When an investigator writes “failure to establish adequate written procedures,” they are constructing Firm-level accountability through systemic failure framing. When they write “operator failed to follow established procedure,” they are constructing Individual-level

accountability through agent-action framing. These linguistic choices are not neutral: they determine who bears responsibility and what corrective actions follow.

Prior work on accountability in computational systems has examined graph-level structures (3) and convergence patterns (1), but has not examined how accountability is linguistically constituted at the token level. I address this gap by treating responsibility attribution as a text classification problem and asking: can responsibility type be predicted from violation text alone, and if so, what linguistic patterns drive each class?

This paper makes three contributions:

- I formalize responsibility attribution as a three-class NLP classification task (Firm / Individual / System) grounded in GMP regulatory linguistics.
- I demonstrate that attribution classes are linguistically distinguishable, with Individual-level violations exhibiting the strongest signal (AUC = 0.940) and System-level the weakest (AUC = 0.764).
- I show that FDA Warning Letter documents systematically over-attribute infrastructure failures to Firm-level responsibility, providing linguistic evidence of accountability convergence.

Related Work

Text Classification for Regulatory Documents

NLP-based classification has been applied to clinical notes (6), legal documents, and compliance text. Recent surveys on relation extraction (5) demonstrate that typed relationships between entities can be extracted from unstructured text. I extend this work to responsibility attribution, treating the subject-violation relationship as a classification target rather than an extraction target.

Interpretable Text Classification

LIME (4) provides local, model-agnostic explanations for text classifiers by identifying which tokens most strongly influence individual predictions. I apply LIME to responsibility attribution to reveal the linguistic markers that constitute each accountability class in regulatory text.

Computational Accountability

Prior work has operationalized accountability as a graph structure (3) and a convergence process (1; 2). I extend this program by examining accountability at the lexical level, treating word choice as the primary mechanism through which responsibility is constructed.

Framework

Task Definition

I define responsibility attribution prediction as a three-class text classification task:

- **Firm**: organizational or systemic failure language — the failure is attributed to the company's quality systems, procedures, or oversight
- **Individual**: agent-action language — the failure is attributed to a specific person's action or omission
- **System**: infrastructure failure language — the failure is attributed to equipment, software, or physical systems

Feature Extraction

I use TF-IDF vectorization with unigram and bigram features (max 200 features, sublinear TF normalization) to represent violation text. This sparse representation preserves interpretability while capturing key lexical patterns.

Classification

I evaluate three classifiers: Logistic Regression (LR), Random Forest (RF), and SVM with RBF kernel. Given the small annotated corpus (n=45), I use Leave-One-Out cross-validation (LOO-CV) for robust performance estimation.

Experiments

Dataset

I construct a manually annotated corpus of GMP regulatory violation language (n=45, 15 per class) based on domain knowledge of FDA enforcement linguistics. Each instance is a violation text phrase annotated with its responsibility subject class. For external validation, I apply the trained model to the FDA Warning Letter corpus (N=50, 160 violations, 2016–2025) (2).

Linguistic Feature Analysis

Figure 1 shows the top TF-IDF features per class. Clear class-specific patterns emerge: Firm violations are characterized by *inadequate*, *failure to*, *procedures*, and *program*; Individual violations by *operator*, *employee*, *analyst*, and *error*; System violations by *system*, *equipment*, *malfunction*, and *calibration*.

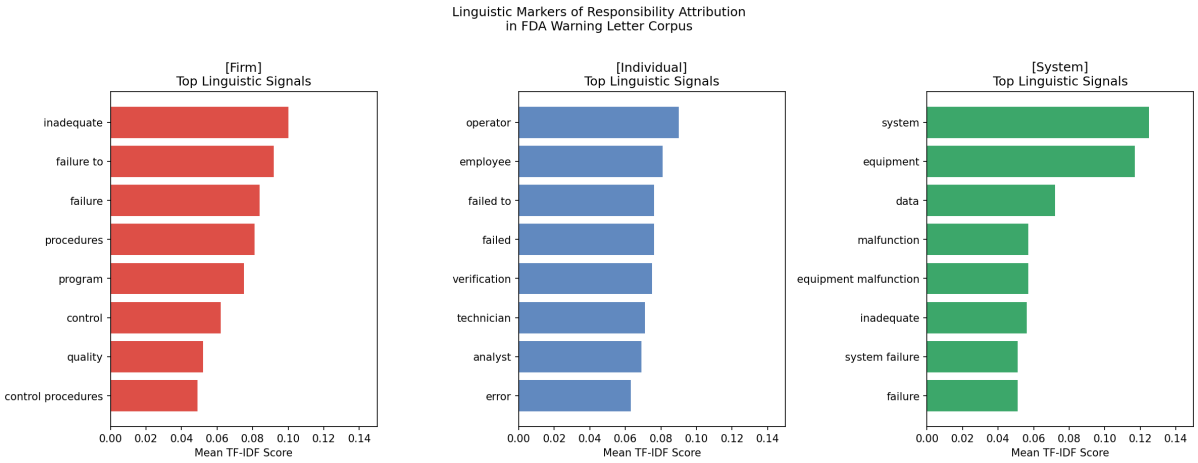


Figure 1: Top TF-IDF features per responsibility class. Firm: systemic failure language. Individual: agent-action language. System: infrastructure failure language.

Classification Results

Table 1 reports LOO-CV performance. Logistic Regression achieves the highest macro F1 (0.725), outperforming Random Forest (0.712) and SVM (0.556). The lower SVM performance reflects overfitting on the small annotated corpus with RBF kernel.

Table 1: LOO-CV Classification Results

Model	**Macro F1**	**Best class**
Logistic Regression	0.725	Individual (0.85)
Random Forest	0.712	Individual (0.83)
SVM	0.556	Individual (0.72)

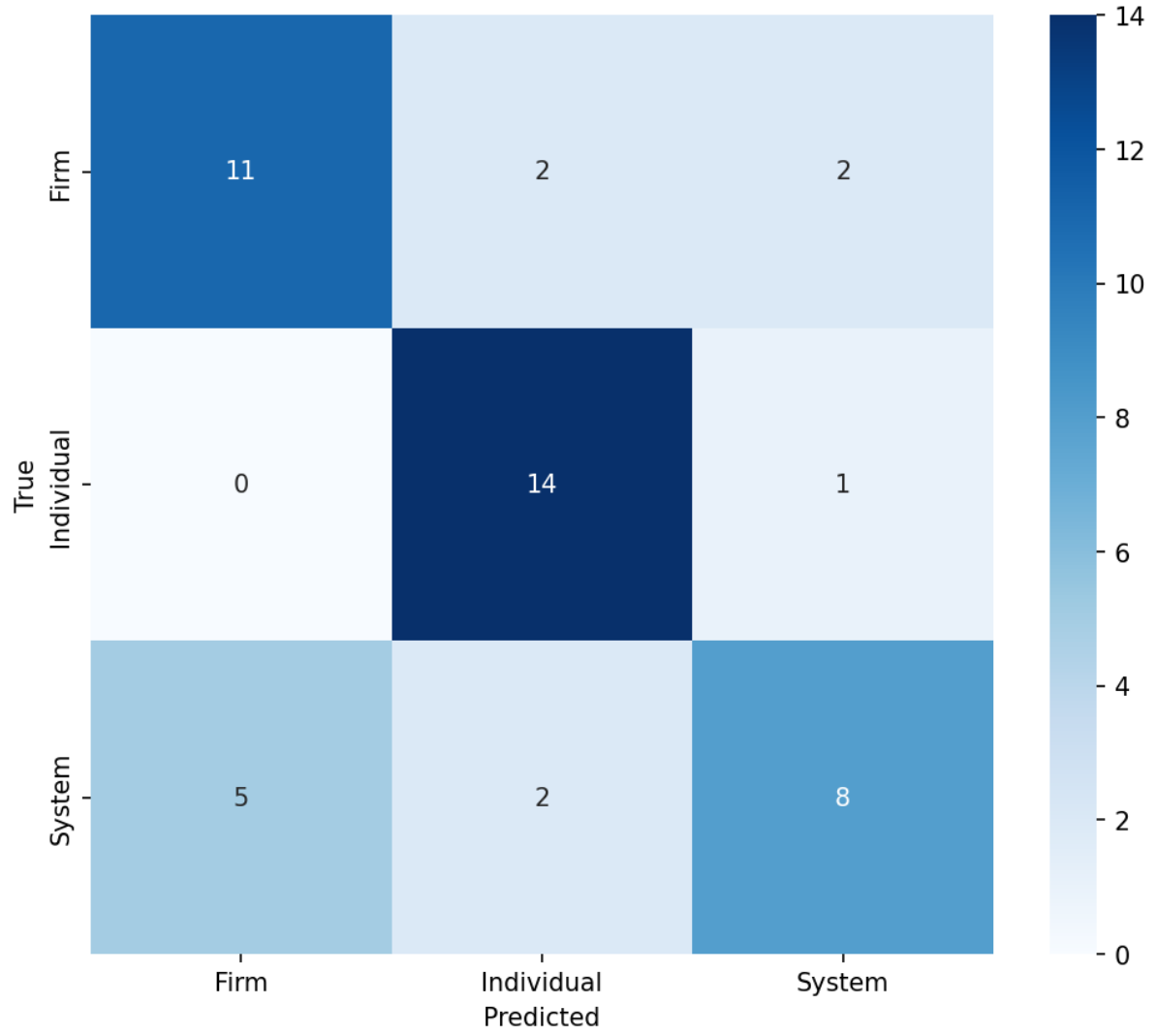


Figure 2: Confusion matrix (LOO-CV, Logistic Regression). System is most frequently confused with Firm.

ROC Analysis

Figure 3 presents per-class ROC curves. Individual-level attribution achieves AUC = 0.940. Firm-level achieves AUC = 0.793, and System-level AUC = 0.764, indicating linguistic overlap between infrastructure and organizational failure language.

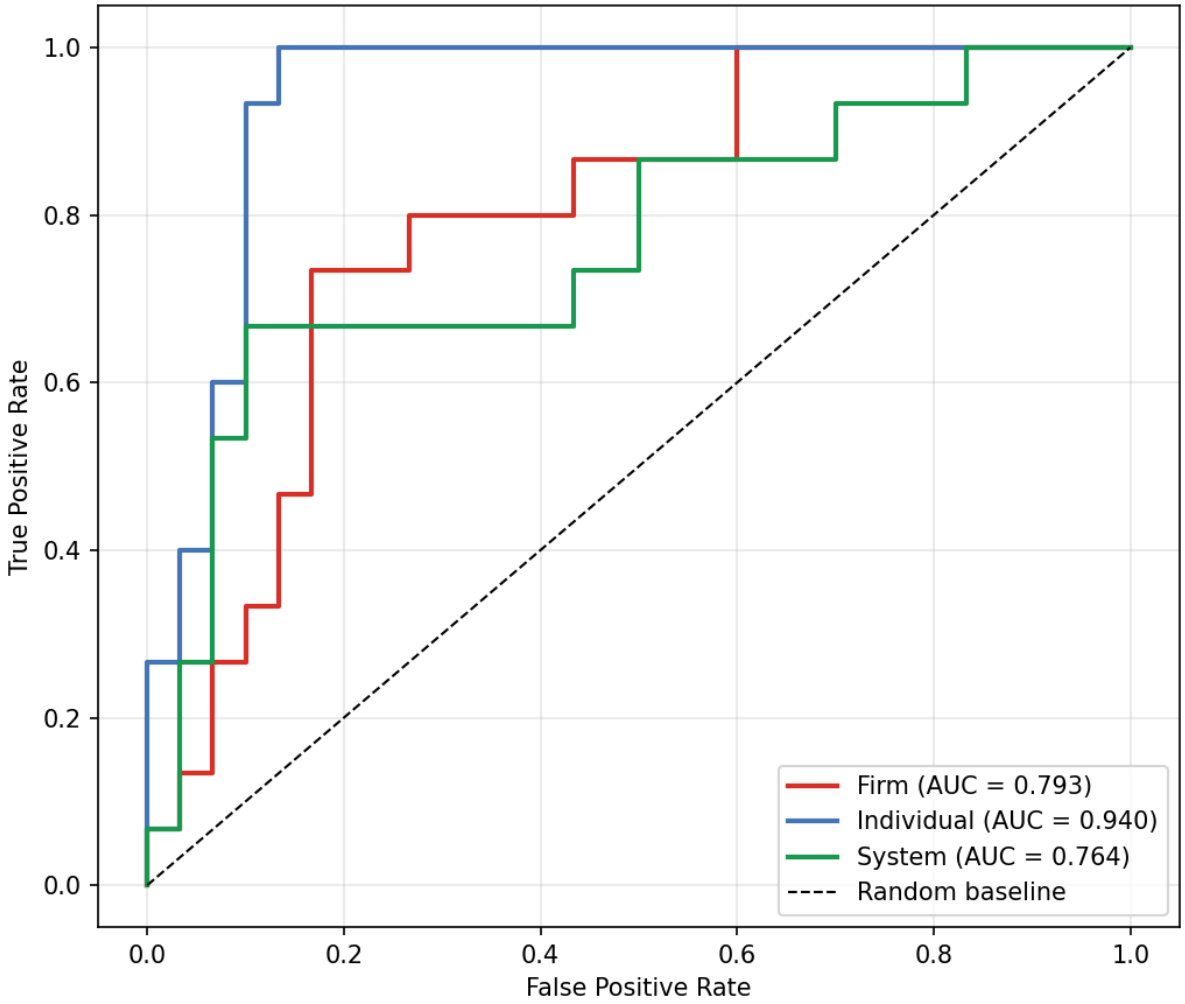


Figure 3: ROC curves per class. Individual achieves AUC = 0.940; System achieves AUC = 0.764.

LIME Analysis

Figure 4 presents LIME explanations for representative instances per class. *procedures* and *failure* are the strongest positive signals for Firm; *operator* alone dominates Individual prediction; *equipment* and *malfunction* dominate System prediction. Notably, *cleaning* carries negative weight for Firm, suggesting that cleaning-related language is associated with System or Individual framing.

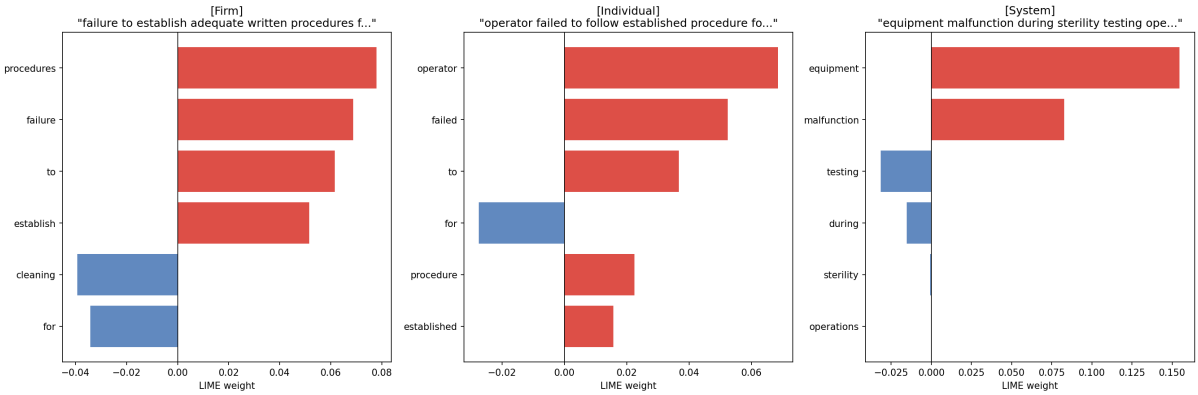


Figure 4: LIME explanations for representative instances. Left: Firm (*procedures*, *failure* dominate). Center: Individual (*operator* dominates). Right: System (*equipment*, *malfunction* dominate).

FDA Corpus Validation

Figure 5 shows that the model reassigns 49 of 160 Firm-labeled violations (30.6%) to System when applied to the FDA Warning Letter corpus. Prediction confidence is uniformly low (0.35–0.65), reflecting linguistic ambiguity between classes in real regulatory text. Figure 6 reveals that violations predicted as System contain infrastructure-specific vocabulary (*equipment*, *system*, *validation*), while Firm-predicted violations contain organizational vocabulary (*quality*, *oversight*, *release*).

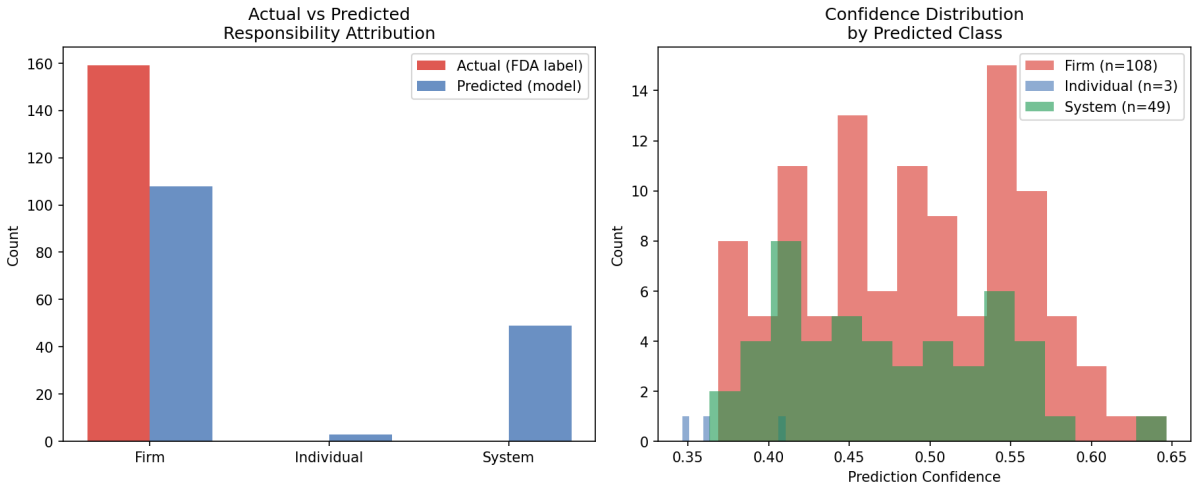


Figure 5: Left: Actual vs predicted class distribution on FDA Warning Letter corpus (49 Firm-labeled violations reassigned to System). Right: Prediction confidence distribution (0.35–0.65), reflecting linguistic ambiguity in real regulatory text.

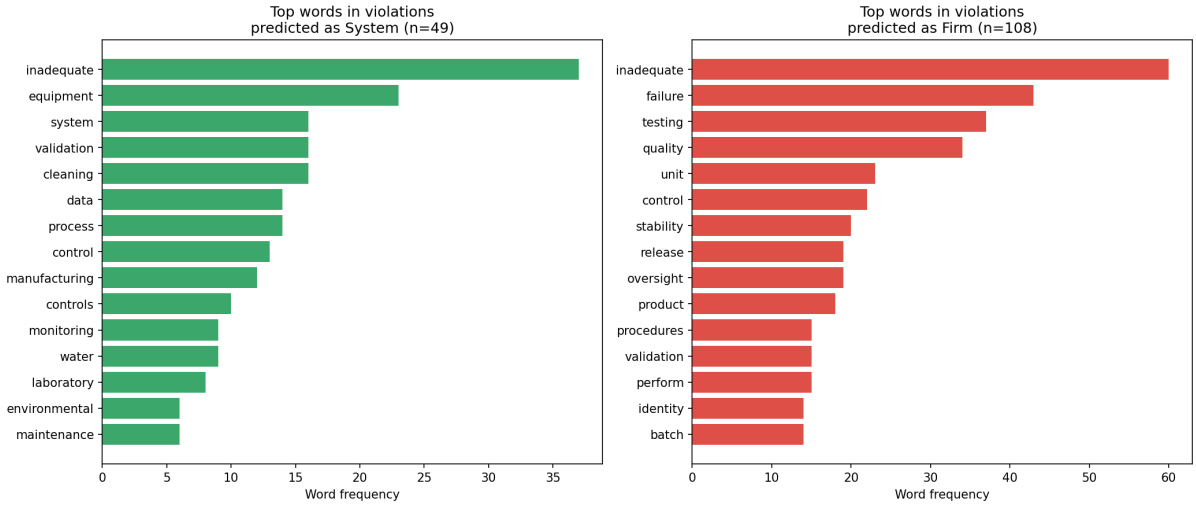


Figure 6: Top words in violations predicted as System (left) and Firm (right). System: infrastructure vocabulary dominates. Firm: organizational vocabulary dominates.

Discussion

Key Findings

Finding 1: Responsibility attribution is linguistically constituted. TF-IDF features alone achieve macro F1 = 0.725, demonstrating that accountability class leaves detectable lexical traces. Attribution is not merely a factual determination — it is constructed through specific word choices that signal organizational, individual, or infrastructural framing.

Finding 2: Individual attribution is most linguistically distinct. Individual-level violations achieve AUC = 0.940, driven by strong agent-noun signals (*operator*, *analyst*, *employee*). Naming a specific role type is the primary mechanism for constructing individual accountability.

Finding 3: System-Firm boundary is linguistically ambiguous. System-level violations are most frequently misclassified as Firm (5/15 in LOO-CV), reflecting genuine linguistic overlap: infrastructure failures are often described using organizational language (*inadequate*, *controls*), making the System-Firm boundary porous.

Finding 4: FDA documents over-attribute infrastructure failures to Firm. The model reassigns 30.6% of Firm-labeled FDA violations to System, with low confidence across all predictions. Regulatory enforcement language systematically collapses System-level failures into Firm-level accountability — a linguistic mechanism consistent with premature accountability convergence (1).

Limitations

Three limitations should be noted. First, the annotated corpus is small (n=45) and domain-specific; generalizability to other regulatory domains requires further validation. Second, the FDA Warning Letter corpus provides no ground truth for System or Individual labels, making external validation of reassignment claims impossible without additional annotation. Third, TF-IDF features do not capture syntactic structure or discourse context; future work should incorporate dependency parsing and transformer-based representations.

Implications

The System-Firm linguistic overlap has direct implications for accountability-aware AI design. Automated classification systems must account for this ambiguity — a violation described as "inadequate equipment cleaning controls" may be Firm-level (inadequate oversight) or System-level (equipment failure) depending on context that surface language alone cannot resolve.

Conclusion

I presented a text-based framework for predicting responsibility attribution from regulatory violation text. The framework demonstrates that accountability class is linguistically constituted — detectable through lexical patterns, yet imperfectly so. Individual-level attribution is most linguistically distinct (AUC = 0.940), while System-level attribution overlaps substantially with Firm-level language (AUC = 0.764). Applied to the FDA Warning Letter corpus, the model identifies 49 violations whose language signals System-level causes despite Firm-level labels, providing linguistic evidence of accountability convergence in regulatory enforcement documents.

Future work will extend this framework to transformer-based models, incorporate syntactic features, and develop a multi-label schema to capture violations with mixed attribution patterns.

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